Physics Department
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FSPP0034 PHYSICS II **Tutorial 6**

Chapter 11-13

1. How hot is metal being welded if it radiates most strongly at 460 nm?

$\lambda_p T = 2.9\times10^{-3}$, $T = \frac{2.9\times10^{-3}}{460\times10^{-9}} = 6304.35$ [Ans: 6300 K]

2. About 0.1 eV is required to break a "hydrogen bond" in a protein molecule. Calculate the minimum frequency and maximum wavelength of a photon that can accomplish this.

$E = hf$, $0.1(1.6\times10^{-19}) = 6.63\times10^{-34}(f)$ $c = f\lambda$

$f = 2.413\times10^{13}$ Hz $\lambda = \frac{3\times10^8}{2.413\times10^{13}} = 1.24\times10^{-5}$ m [Ans: 2 x 10^{13} Hz, 1.24 x 10^{-5} m]

3. Calculate the momentum of a photon of yellow light of wavelength 6.20 x 10^{-7} m.

$p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda} = \frac{6.63\times10^{-34}}{6.20\times10^{-7}} = 1.07\times10^{-27}$ kg m/s [Ans: 1.07 x 10^{-27} kg m/s]

4. What wavelength photon would have the same energy as a 145-gram baseball moving 30.0 m/s?

$KE = E = \frac{hc}{\lambda}$

$KE = \frac{1}{2}mv^2 = \frac{1}{2}(0.145)(30^2) = 65.25$ J $65.25 = \frac{6.63\times10^{-34}(3\times10^8)}{\lambda}$, $\lambda = 3.048\times10^{-27}$ m [Ans: 3.05 x 10^{-27} m]

5. When UV light of wavelength 285 nm falls on a metal surface, the maximum kinetic energy of emitted electrons is 1.70 eV. What is the work function of the metal?

$E = K_{max} + W_0$, $\frac{hc}{\lambda} = 1.70\times1.6\times10^{-19} + W_0$, $W_0 = 4.26\times10^{-19}$ J $= 2.662$ eV [Ans: 2.65 eV]

6. Calculate the wavelength of a 0.23-kg ball traveling at 0.10 m/s.

$KE = \frac{1}{2}mv^2$ $1.15\times10^{-3} = \frac{hc}{\lambda}$ [Ans: 2.9 x 10^{-32} m]

$= \frac{1}{2}(0.23)(0.10)^2 = 1.15\times10^{-3}$ $\lambda = 1.73\times10^{-22}$

7. Determine the binding energy of the last neutron in a $^{32}_{15}$P nucleus. Mass of $^{32}_{15}$P = 31.973907 u.

$^{31}_{15}P + ^{1}_{0}n \rightarrow ^{32}_{15}P$ $\Delta m = (m_{p31} + m_n) - m_{p32}$

$E = \Delta mc^2$ $= (30.973762 + 1.00866) - 31.973907$ [Ans: 7.94 MeV]

$= 1.272\times10^{-12}$ J $= 7.95$ MeV $= 0.008515$ u $= 1.413\times10^{-29}$ kg

8. Calculate the total binding energy, and the binding energy per nucleon, for $^{7}_{3}$Li . Given: mass of $^{7}_{3}$Li = 7.016005 u .

$E = \Delta mc^2 = (6.720178\times10^{-29})(3\times10^8)^2$ $\frac{37.8}{7} = 5.4$ MeV/n

$\Delta m = (Zm_p + Nm_n) - m_A$ $= 6.04816\times10^{-12}$ J $= 37.8$ MeV [Ans: 5.61 MeV/nucleoun]

$= (3\times1.007276 + 4\times1.008665) - 7.016005 = 0.040483$ u $= 6.720178\times10^{-29}$ kg

9.. How much energy is released when tritium, $^{3}_{1}$H decays β^- by emission? Mass of $^{3}_{1}$H = 3.016049 u and $^{3}_{2}$He = 3.016029 u.

$^{3}_{1}H \rightarrow ^{3}_{2}He + ^{0}_{-1}e$

$Q = (m_p - m_d)c^2 = (3.016049 - 3.016029)(1.66\times10^{-27})(3\times10^8)^2$ [Ans: 0.019 MeV]

$= 2.988\times10^{-15}$ J $= 0.018675$ MeV

10. The isotope $^{218}_{84}$Po can decay by either α or β emission. What is the energy release in each case? The mass of $^{218}_{84}$Po is 218.008965 u.

$^{218}_{84}Po \rightarrow ^{214}_{82}Pb + \alpha$ $Q = (m_p - m_d)c^2$ $^{218}_{84}Po \rightarrow ^{218}_{85}At + ^{0}_{-1}e$

$Q = m_pc^2 - (m_d + m_\alpha)c^2$ $= (218.008965 - 218.008694)c^2$ $= 4.04874\times10^{-14} = 0.253$ MeV

$= 218.008965(3\times10^8)^2(1.66\times10^{-27}) - (213.999805 + 4.002603)(3\times10^8)^2(1.66\times10^{-27})$ [Ans: α: 6.108 MeV , β = 0.252 MeV]

$= 3.2570553937\times10^{-8} - 3.256955976\times10^{-8} = 9.7961\times10^{-13}$ J $= 6.123$ MeV

11. What is the decay constant of $^{238}_{92}$U whose half-life is 4.5 x 10^9 year?

$T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$, $1.39968\times10^{10} = \frac{\ln(2)}{\lambda}$, $\lambda = 4.952\times10^{-18}$ s^{-1} [Ans: 4.9 x 10^{-18} s^{-1}]

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4.5×10^9 years $= 5.4\times10^{10}$ months $= 1.39968\times10^{17}$ s half life =

12. What is the activity of a sample of $^{14}_{6}C$ that contains 8.1 x 10^{20} nuclei?

$14g \to 6.023\times10^{23}$ nuclei
$xg \to 8.1\times10^{20}$ nuclei
$x = 0.018828\,g$
$T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}$
$\lambda = 3.889\times10^{-12}$
$A = N\lambda = 3.1501\times10^{9}$ d/s

[Ams: 3.1 x 10^9 decays/s]

13. The activity of a radioactive source decreases by 2.5% in 31.0 hours. What is the half-life of this source?

$A = 0.975A_0$
$0.975A_0 = A_0 e^{-\lambda t}$
$0.975A_0 = A_0 e^{-1.116\times10^{5}\lambda}$
$\ln(0.975) = -1.116\times10^{5}\lambda$
$\lambda = 2.2686\times10^{-7}\,s^{-1}$
$T_{\frac{1}{2}} = \frac{\ln(2)}{\lambda} = 3055396\,s = 35.36$ days

[Ans: 35.4 days]

14. Determine whether the reaction $^{2}_{1}H + ^{2}_{1}H \rightarrow ^{3}_{2}He + n$ requires a threshold energy.

[Ans: No]

15. (a) Can the reaction occur if the $p + ^{7}_{3}Li \rightarrow ^{4}_{2}He + \alpha$ occur if incident proton has kinetic energy = 3500 keV?

(b) If so, what is the total kinetic energy of the products? If not, what kinetic energy is needed?

[Ans: (a) Occur (b) 20.8 MeV]

16. What is the energy released in the fission reaction of

$n + ^{235}_{92}U \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3n$. (The masses of $^{141}_{56}Ba$ and $^{97}_{36}Kr$ are 140.914411 u and 91.926156 u, respectively.)

[Ans:173.3 MeV]

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